



Summary of knowledge sessions as on 17/03/2026

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This page curates the ML/DL-oriented concepts covered across recent knowledge sessions and technical discussions. The focus is on preserving the original intent and terminology, while organizing the content into clear, reusable notes.

Overfitting (concept + intuition) — 23/01/2026

Overfitting was discussed through the lens of the end-to-end ML training workflow. The central idea was that a model can show strong performance on data it has already seen, while failing to generalize to unseen examples.

Key framing:

- Overfitting is not limited to deep learning. The same pattern appears in classical regression models and modern LLM-based systems.
 - Evaluation should reflect generalization behavior, not memorization of training distributions.
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Loss functions for classification — 27/01/2026

This session focused on what loss functions do in supervised learning and how they connect to optimization.

Core concept:

- A loss function provides the training objective.
- Optimization algorithms iteratively adjust parameters to reduce (or, in some formulations, maximize) the objective.

Practical intuition emphasized:

- “Fine-tuning” can be understood as shaping what the model learns by changing the strength of the learning signal.
- Weighting examples changes which distinctions the model is incentivized to learn.
 - Example discussed: improving a “wolf vs dog” classifier by adjusting the effective sample mix so the model receives a stronger learning signal for the class that needs improvement.

Takeaway:

- A strong engineering intuition often requires converting informal reasoning (“this error matters more”) into measurable objectives.

Loss functions for regression (MAE, MSE, RMSE) — 28/01/2026

This discussion connected loss-function selection to real-world interpretability.

Core concept:

- Different losses represent different definitions of “error,” and those choices change how a model fits data.

Progression explained:

- A naïve error term like $(\text{pred} - \text{actual})$ can cancel out positive and negative deviations.
- MAE avoids cancellation via absolute values.
- MSE is commonly used because it is smooth and convenient for optimization, but it emphasizes larger errors.
- RMSE is used when it is important that the error metric remain in the original units of the target variable.

Practical framing:

- A cement-strength example was used to highlight that “what error should mean” depends on the real system, not only mathematical convenience.

Generalization, capacity, bias–variance, and data leakage — 03/02/2026

This session framed generalization as the primary objective of ML systems: performing reliably on data the model has not previously encountered.

Key concepts:

- **Model capacity:** the complexity of the function class the model can represent.
- **Bias–variance tradeoff:**
 - Lower-capacity models can be stable and interpretable, but may underfit.
 - Higher-capacity models can fit richer patterns, but may be more sensitive to variance and harder to explain.
- **Non-linearity** increases modeling capacity while increasing complexity.

Data leakage was highlighted as a major risk:

- Exposing test information (or test-like signals) during development can inflate apparent performance without improving real reliability.

Practical takeaway:

- Prefer a model that is reliable and stable over one that looks strong only due to leakage or overfitting.

Why accuracy is not preferred for classification (confusion matrix, precision/recall, ROC) — 04/02/2026

This section addressed why accuracy can be misleading in classification, particularly under class imbalance.

Core points:

- The **confusion matrix** gives a more complete view of error modes than a single aggregate metric.
- **Precision** answers: “When the model predicts positive, how often is it correct?”
- **Recall** answers: “Out of the true positives, how many were recovered?”

- **ROC** visualizes the tradeoff between true positive rate and false positive rate across thresholds.
- **AUC** summarizes separability across thresholds.

Engineering takeaway:

- Metric selection should reflect product constraints and real failure modes, not just a single headline number.
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Building ML intuition from first principles (CNN → neural nets → model evolution) — 20/01/2026

This session emphasized developing “why-based” understanding rather than only implementation knowledge.

Key narrative:

- It is often possible to explain *how* a CNN works at a code level.
- The deeper skill is understanding *why* the architecture evolved and what each change solved.

Mental model captured:

- Linear regression → logistic regression → stacking logistic units → neural networks.

Takeaway:

- Understanding motivations behind architecture evolution improves reasoning about what improvements are likely to matter and why.
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Maximum Likelihood Estimation (MLE) as a foundation — 24/02/2026

MLE was described as a foundational idea underlying modern statistical learning.

Core concept:

- Parameters are chosen to maximize the likelihood of the observed data under the model.

Supporting idea referenced:

- The discussion connected this to broader statistical foundations (including CLT-related intuition for aggregated random effects).

Takeaway:

- A clear grasp of MLE helps with understanding estimation, training objectives, and why many optimization problems are framed the way they are.
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Multi-modal models, attention, U-Net, and video generation (LTX2 + control via LoRA) — 11/03/2026

This session focused on how architectures evolved toward attention-centric systems, and how these ideas extend into multi-modal generation.

Concepts discussed:

- **Attention** as a mechanism that improved earlier architectures (including CNN-based systems).
- **U-Net** as a downscale–upscale pattern designed to preserve multi-scale structure, widely used in generation/segmentation.
- A broader trend: attention mechanisms improved performance across major model families, eventually leading to architectures dominated by attention.

Multi-modal framing:

- Multi-modal models are trained on paired modalities such as text–video and text–audio.
- Outputs can be guided via control signals.

LoRA-related note (high level):

- LoRA components can function as “control” mechanisms.
- In-context LoRA was mentioned as a related variant with distinct benefits.

Suggested learning path captured:

- Start with transformer fundamentals.
 - Study the LTX2 repository and architecture.
 - Expand to Stable Diffusion and the broader ecosystem.
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Fine-tuning: LoRA and QLoRA + practical workflow considerations — 12/03/2026

This session consolidated fine-tuning into an engineering-focused understanding.

Core concepts:

- **LoRA**: adapts a model using trainable low-rank matrices while keeping the base weights frozen.
- **Rank**: a control knob for the parameter footprint and adaptation capacity.
- **QLoRA**: applies quantization before LoRA to reduce compute and memory cost.

Proposed workflow (high level):

- Select an open-source model → train using Unsloth → produce a fine-tuned model → host via Ollama.

Open questions explicitly noted:

- How to evaluate fine-tuned performance for the target use case.
- How to obtain a dataset of sufficient quality.
- When fine-tuning is warranted versus alternative approaches.

Use cases discussed (high level):

- Domain-specific adaptation, such as improving summarization for medical literature.
- Improving a weaker small model (SLM) that underperforms on a target dataset.